1. Which component of cloud architecture is considered the "engine room" that processes and stores data?

- a. The Front-End
- b. The Client
- c. The Back-End
- d. The Connector

2. What is the ability of a system to withstand hardware or software failures without significant downtime?

- a. Scalability
- b. Elasticity
- c. Performance
- d. Resilience (Fault Tolerance)

3. Which architectural pattern involves distributing incoming traffic across multiple resources to prevent bottlenecks?

- a. Caching
- b. Decoupling
- c. Load Balancing
- d. Microservices

4. In the Well-Architected Framework, which pillar focuses on protecting data and implementing the principle of least privilege?

- a. Operational Excellence
- b. Security
- c. Reliability
- d. Performance Efficiency

5. Networking is often referred to as the __________ of cloud computing.

- a. Front-End
- b. Invisible Backbone
- c. Virtual Machine
- d. Storage Layer

6. Cloud networking ensures connectivity between which three critical points?

- a. Users, Servers, and Storage
- b. Users, Applications, and Data Centers
- c. Developers, Clients, and Routers
- d. CPUs, RAM, and Operating Systems

7. Which protocol is used to protect data in transit within the cloud?

- a. IP Addressing
- b. Routing
- c. VPNs and Firewalls
- d. Load Balancing

8. Moving a virtual machine from one physical host to another without downtime is known as:

- a. Load Balancing
- b. Resource Allocation
- c. Live Migration
- d. Scaling

9. Which area of resource management involves assigning processor time to virtual machines?

- a. CPU Scheduling
- b. Traffic Shaping
- c. Deduplication
- d. I/O Optimization

10. Which type of cloud storage is best suited for databases and virtual machines due to its high performance?

- a. File Storage
- b. Block Storage
- c. Object Storage
- d. Cold Storage

11. What is a key characteristic of Object Storage?

- a. It stores data as files and folders.
- b. It is limited by local hardware.
- c. It stores data as objects with metadata and is highly scalable.
- d. It requires manual backups by the user.

12. Which of the following is considered a limitation or risk of cloud storage?

- a. High upfront cost
- b. User-managed maintenance
- c. Internet dependency
- d. Manual data replication

13. In the context of cloud storage security, what measure ensures data is protected while being moved?

- a. Encryption at rest
- b. Encryption in transit
- c. Multi-factor authentication
- d. Data deduplication

14. Which cloud storage service is an example of File Storage?

- a. Amazon EBS
- b. Amazon S3
- c. Google Drive
- d. Azure Blob Storage

15. What is the "pay-as-you-go" model in cloud storage an alternative to?

- a. Automatic backups
- b. High upfront costs of local storage
- c. Scalability
- d. Internet dependency

16. Who is responsible for the physical security of data centers in the Shared Responsibility Model?

- a. The User
- b. The Cloud Provider
- c. Both the User and Provider
- d. The Internet Service Provider

17. Which of these is a customer's responsibility in the Shared Responsibility Model?

- a. Infrastructure security
- b. Availability and resilience
- c. Identity and Access Management (IAM)
- d. Physical hardware maintenance

18. What does the "Least Privilege Principle" refer to in cloud security?

- a. Using the same password for all accounts.
- b. Limiting user privileges to only what is necessary for their job.
- c. Allowing all employees to access the root account.
- d. Disabling firewalls for faster access.

19. A Database Appliance is a "DIY" system where the user must manually source and integrate servers, storage, and switches.

- a. True
- b. False

20. "Capacity-on-Demand" allows a business to license fewer CPU cores initially and "turn on" more cores via software as they grow.

- a. True
- b. False

21. Which mechanism is used to protect a network by controlling incoming and outgoing traffic?

- a. Data Encryption
- b. Firewall
- c. Role-Based Access Control
- d. Multi-Factor Authentication (MFA)

22. According to the "Pizza as a Service" analogy, which model is like "Dining Out" where everything is managed for you?

- a. IaaS
- b. PaaS
- c. SaaS
- d. On-premises

23. What is a key benefit of using a Cloud Service Provider instead of buying hardware?

- a. Manual maintenance of physical servers.
- b. No need to buy expensive hardware; you "pay-as-you-go."
- c. Limited scalability.
- d. Total control over physical data center cooling.

24. What is a Database Appliance?

- a. A software-only solution for managing databases.
- b. An "all-in-one" platform pre-integrated with hardware, software, networking, and storage.
- c. A DIY server build where users match their own hardware.
- d. A cloud-only service with no physical component.

25. What does "Capacity-on-Demand" licensing allow users to do?

- a. Pay for all CPU cores upfront.
- b. Start with a few licensed cores and "turn on" more via software as needed.
- c. Replace hardware every time more power is required.
- d. Use the database for free during off-peak hours.

26. Which feature of a DB Appliance reduces the risk of bugs?

- a. Using different hardware for every user.
- b. The "Single Patch" experience where the vendor tests all updates together.
- c. Letting users choose their own OS and firmware versions.
- d. Avoiding all software updates.

27. How long does it typically take to deploy a production-ready database using an appliance?

- a. 30-60 minutes
- b. 3-5 days
- c. 2-4 weeks
- d. 6 months

28. In cloud-native development, what is the preferred method for scaling?

- a. Vertical scaling (buying a bigger server).
- b. Horizontal scaling (adding more small servers).
- c. Manual scaling during off-hours.
- d. Reducing the number of users.

29. Which tool is used for "Orchestration" to manage Docker containers?

- a. Terraform
- b. Kubernetes
- c. GitHub
- d. Python

30. What does "Infrastructure as Code" (IaC) allow developers to do?

- a. Write code manually on every server.
- b. Use scripts (like Terraform) to build servers and databases automatically.
- c. Eliminate the need for any cloud infrastructure.
- d. Replace the CI/CD pipeline.

31. Which tool is used for "Observability" to monitor traces and logs in cloud apps?

- a. Jenkins
- b. Datadog or New Relic
- c. Docker
- d. Go

32. In a Public Cloud deployment model, the infrastructure is used exclusively by a single organization.

- a. True
- b. False

33. Platform as a Service (PaaS) provides a framework for developers to build, test, and deploy applications without managing the underlying servers.

- a. True
- b. False

34. The "Front-End" of cloud architecture refers to the hardware and software located at the cloud provider's data center.

- a. True
- b. False

35. Scalability refers to the ability of a system to handle a growing amount of work by adding resources.

- a. True
- b. False

36. A "Hybrid Cloud" combines elements of both public and private clouds, allowing data and applications to be shared between them.

- a. True
- b. False

37. Which cloud deployment model is used exclusively by a single organization for security or compliance?

- a. Public Cloud
- b. Private Cloud
- c. Hybrid Cloud
- d. Community Cloud

38. If a startup rents virtual servers like AWS EC2 to host their website, which service model are they using?

- a. SaaS
- b. PaaS
- c. IaaS
- d. FaaS

39. Which service model allows developers to build and deploy applications without worrying about managing the underlying server infrastructure?

- a. Infrastructure as a Service (IaaS)

- b. Platform as a Service (PaaS)
- c. Software as a Service (SaaS)
- d. Desktop as a Service (DaaS)

40. What is a primary benefit of cloud computing over on-premises infrastructure regarding scalability?

- a. On-premises is easier to scale down.
- b. Cloud allows for faster scaling up or down based on use.
- c. Cloud requires more physical space for servers.
- d. Scaling on-premises leads to fewer financial losses.